Exam 1

Statistics (統計學, 專班) · Fall 2026 (115-1) · National Yang Ming Chiao Tung University
Prof. Huei-Wen Teng · 17 October 2026

Name 姓名: _____ Student ID 學號: _____

Time	150 minutes
Coverage	Chapters 1 to 9
Questions	25 multiple-choice questions, 4 points each, 100 points in total
Allowed	One A4 formula sheet (your own), a calculator
Not allowed	Phones, computers, AI tools, books, talking

Instructions

1. Each question has **exactly one** correct answer.
2. Write your answers on the **answer sheet**. Only the answer sheet is graded.
3. There is no penalty for a wrong answer. Answer every question.
4. Textbook tables are on the last two pages: Table 3, areas under the normal curve (negative z, then positive z).
5. You may write on this paper. Hand in the paper and the answer sheet together.

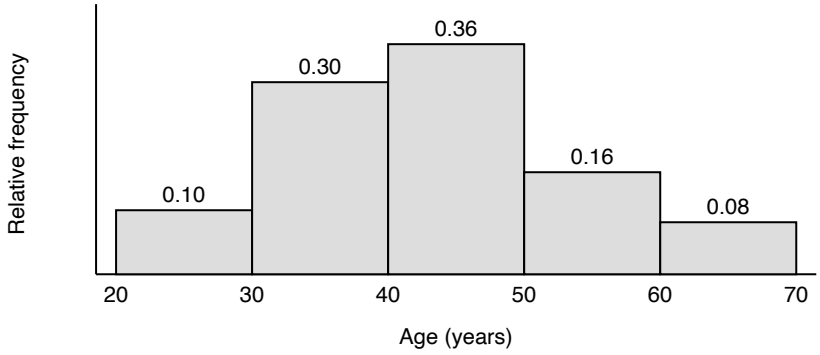
How to use your time

1. Part A (Questions 1–20): about 40 minutes.
2. Part B (Questions 21–25): about 20 minutes.
3. Then check your work. You have plenty of time: do not rush, and do not leave early without checking.

Part A · Questions 1–20 · 4 points each (80 points)

Suggested time: about 40 minutes.

1. A bank records the age of 50 customers who opened an account last month. The relative frequency histogram is shown below. How many of the 50 customers are **younger than 40**?



(A) 30

(B) 15

(C) 20

(D) 40

2. Five customers wait 2, 4, 4, 6 and 9 minutes. What are the sample mean and the sample standard deviation?

(A) mean = 5, s = 2.65

(B) mean = 4, s = 2.65

(C) mean = 5, s = 2.37

(D) mean = 5, s = 7

3. Eleven deposits (in NT$ thousand), in order: 3, 5, 6, 8, 9, 10, 12, 13, 15, 18, 30. What are the median, the lower quartile Q1 and the upper quartile Q3?

(A) median = 10, Q1 = 6, Q3 = 15

(B) median = 11.7, Q1 = 6, Q3 = 15

(C) median = 10, Q1 = 5, Q3 = 18

(D) median = 9, Q1 = 6, Q3 = 15

4. For the monthly credit-card bills of a customer, Q1 = NT$240 and Q3 = NT$440. Using the box plot rule (fences at 1.5 × IQR), which bill is an outlier?

(A) NT$760

(B) NT$700

(C) NT$730

(D) NT$650

5. A bank asks 200 customers whether they use the mobile app.

	Uses the app	Does not use the app	Total
Under 40	90	30	120
40 and over	40	40	80
Total	130	70	200

Among the customers **aged 40 and over**, what proportion use the app?

(A) 0.20

(B) 0.65

(C) 0.31

(D) 0.50

6. For 10 months, a bank records the loan interest rate x and the number of loan applications y. The correlation coefficient is $r = -0.85$. Which statement is correct?

(A) Months with higher rates tend to have more applications, and the linear relationship is strong.

(B) When the rate rises by 1 point, applications fall by exactly 0.85.

(C) The linear relationship is weak, because r is negative.

(D) Months with higher rates tend to have fewer applications, and the linear relationship is strong.

7. A customer answers three yes/no questions: Do you have a credit card? Do you use the mobile app? Do you have a mortgage? How many simple events are in the sample space?

(A) 6

(B) 9

(C) 8

(D) 3

8. Five loan files A, B, C, D and E are on a desk. You take one file at random, and then your colleague takes one of the four that are left. What is the probability that you get file E and your colleague does **not** get file A?

(A) 0.12

(B) 0.15

(C) 0.30

(D) 0.20

9. Let x be the number of new accounts opened at a branch in one hour.

x	0	1	2	3
p(x)	0.1	0.4	0.3	0.2

What is the expected value of x?

(A) 1.5

(B) 1.6

(C) 2.0

(D) 1.0

10. Each of 4 customers accepts an offer with probability 0.5, independently. What is the probability that **exactly 2** accept?

(A) 0.250

(B) 0.0625

(C) 0.500

(D) 0.375

11. The time a customer waits for a teller is uniformly distributed from 0 to 20 minutes. What is the probability that a customer waits **more than 15 minutes**?

(A) 0.75

(B) 0.05

(C) 0.25

(D) 0.15

12. The daily cash withdrawn from an ATM is normally distributed with mean NT$500 thousand and standard deviation NT$50 thousand. What is the probability that the withdrawals on a day are **more than NT$600 thousand**?

(A) 0.0228

(B) 0.9772

(C) 0.0456

(D) 0.4772

13. Half of a bank's customers hold its credit card ($p = 0.5$). In a random sample of 100 customers, x hold the card. Use the normal approximation **with the continuity correction**. What is $P(x \geq 60)$?

(A) 0.0179

(B) 0.9713

(C) 0.0228

(D) 0.0287

14. A population has N = 4 numbers: 1, 3, 5, 7. A random sample of n = 2 is taken **without replacement**. What is the mean of the sampling distribution of the sample mean $\bar{x}$?

(A) 6

(B) 2

(C) 4

(D) 16

15. A random sample of n = 36 is taken from a population with mean $\mu = 50$ and standard deviation $\sigma = 12$. What are the mean and the standard error of the sample mean $\bar{x}$?

(A) mean = 50, standard error = 12

(B) mean = 50, standard error = 2

(C) mean = 50, standard error = 0.33

(D) mean = 8.33, standard error = 2

16. In a large population, 20% of customers pay their bill late ($p = 0.2$). For a random sample of n = 100 customers, what are the mean and the standard error of the sample proportion $\hat{p}$?

(A) mean = 0.20, standard error = 0.40

(B) mean = 20, standard error = 4

(C) mean = 0.20, standard error = 0.04

(D) mean = 0.20, standard error = 0.0016

17. A random sample of n = 100 transactions has mean $\bar{x}$ = NT$30 thousand and standard deviation s = NT$5 thousand. What is the 95% margin of error for estimating the population mean?

(A) 0.82

(B) 0.98

(C) 0.50

(D) 9.80

18. A random sample of n = 64 customers has a mean waiting time of $\bar{x} = 8.0$ minutes with standard deviation s = 2.0 minutes. What is the 95% confidence interval for the population mean waiting time?

(A) (7.59, 8.41)

(B) (4.08, 11.92)

(C) (7.51, 8.49)

(D) (7.75, 8.25)

19. A bank says the mean waiting time is 5 minutes. A manager believes it is **longer**. A random sample of n = 100 customers gives $\bar{x} = 5.3$ minutes and s = 1.5 minutes. Which hypotheses and test statistic are correct?

(A) $H_0: \mu = 5$, $H_a: \mu > 5$, $z = 2.0$

(B) $H_0: \mu = 5$, $H_a: \mu > 5$, $z = 0.2$

(C) $H_0: \mu = 5$, $H_a: \mu \neq 5$, $z = 2.0$

(D) $H_0: \mu > 5$, $H_a: \mu = 5$, $z = 2.0$

20. A bank claims that the mean time to process a loan is 10 days. To check the claim, an auditor takes a random sample of $n = 64$ loans and finds $\bar{x} = 10.6$ days and $s = 2.0$ days. Test H_0: $\mu = 10$ against H_a: $\mu \neq 10$ with $\alpha = 0.05$. What is the conclusion?

(A) Do not reject H_0, because $z = 0.3$ is less than 1.96.

(B) Do not reject H_0, because 10.6 is close to 10.

(C) Do not reject H_0, because $z = 2.4$ is not greater than 2.58.

(D) Reject H_0, because $z = 2.4$ is greater than 1.96.

Part B · Questions 21–25 · 4 points each (20 points)

Suggested time: about 20 minutes. These questions need more than one step.

21. Each loan defaults with probability 0.2, independently. A loan officer has 5 loans. Let x be the number that default. Which line is correct?

(A) P(x = 1) = 0.328, mean = 1, standard deviation = 0.894

(B) P(x = 1) = 0.205, mean = 1, standard deviation = 0.894

(C) P(x = 1) = 0.410, mean = 1, standard deviation = 0.800

(D) P(x = 1) = 0.410, mean = 1, standard deviation = 0.894

22. Credit scores are normally distributed with mean 650 and standard deviation 50. The bank gives its best rate to the customers in the **top 10%** of scores. What is the lowest score that gets the best rate?

(A) 732

(B) 714

(C) 700

(D) 586

23. Card transactions have mean NT$1,200 and standard deviation NT$400. The distribution is skewed. For a random sample of n = 100 transactions, what is the probability that the sample mean is **between NT$1,160 and NT$1,240**?

(A) 0.3413

(B) 0.6826

(C) 0.0796

(D) 0.8413

24. In a random sample of n = 400 customers, 80 say they would move to another bank ($\hat{p}$ = 0.20). (i) Find the 95% confidence interval for p. (ii) The bank now wants a margin of error of 0.02. Using 0.20 as the planning value for p, how many customers must it sample?

(A) (0.161, 0.239); n = 2,401

(B) (0.180, 0.220); n = 1,537

(C) (0.161, 0.239); n = 1,537

(D) (0.161, 0.239); n = 800

25. In the past, 20% of small-business loans defaulted in the first year. The bank starts a new screening rule and hopes that the default rate is now **lower**. In a random sample of n = 400 loans under the new rule, 60 default. Use $\alpha = 0.05$. Which line is correct?

(A) z = –2.50, p-value = 0.0062; there is evidence that the default rate is lower.

(B) z = –2.50, p-value = 0.0124; there is evidence that the default rate is lower.

(C) z = –2.50, p-value = 0.9938; there is no evidence that the default rate is lower.

(D) z = –2.80, p-value = 0.0026; there is evidence that the default rate is lower.

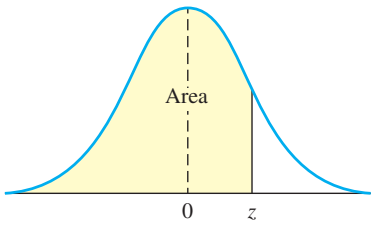


Table 3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

■ **Table 3** (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998